Adi Atrash

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Summary

An avid aerospace enthusiast and second year aerospace student at the Technion, I am interested in the fields of fluid/aero dynamics, propulsion, CFD and robotics.

Education

Technion – Israel Institute of Technology 2024-2028

BSc, Aerospace, aeronautical and astronautical Engineering – avg of 88.5, Deans list

Experience

Aerodynamics lead - Formula student Technion

Jan 2025 – present

- Led a team of 15 students during the aerodynamic design of the race car, optimizing maximum downforce and minimum drag.
- Verified the design using CFD software (**STAR-CCM+ & Ansys Fluent**), optimized the design using FEA tools (**SolidWorks Simulation** & **Altair Inspire).**
- Performed scale resolving simulations to study unsteady effects, using **LES** and **DES** simulations.
- Implemented new CFD methods, like cornering simulations, simulation workflow automation and **adjoint based adaptive mesh refinement.**
- Performed **wind tunnel** testing to validate CFD simulation results.
- Performed **conjugate heat transfer** simulations on electronic components (like the HV battery accumulator).
- Managed the team's **3D printing, Fiberglass** and **carbon fiber composite manufacturing.**
- Contributed towards the development of new composite manufacturing methods aiming to reduce waste and maximize mechanical performance, turning the innovation into business plan idea (#3 FS France 2025).
- Secured funding and multiple sponsorships from local and international industrial partners allowing the team to explore new capabilities.

Research assistant – Technion reactive thermofluidics lab

Jul 2025 – present

- Conducted GPU accelerated CFD simulations on DDT experiments and rotational detonation engines (RDE).
- Designed, fabricated and tested mechanical components for experiments on deflagration to detonation (DDT) and cool-flame phenomena.
- Assisted with experimental setup, instrumentation and data acquisition for thermofluidic studies.

Propulsion engineer - Technion Rocketry Club

Jan 2025 – Apr 2026

- Designed and analyzed LOX tanks for cryogenic operation in a blowdown regime.
- Contributed to the atomizer design, to ensure even fuel-oxidizer mixing for consistent combustion.
- Researched insulation and fail-safe methods, to ensure safety and compatibility with the local conditions.
- Developed an analytical model to describe rocket propulsion performance in a blow down regime.

FIRST FRC

Mentor Jul 2023 – Jul 2025

- An international robotics competition for high school students. I serve as a CAD mentor, helping students realise their ideas using SolidWorks and Onshape.

Student Jul 2018 – Jun 2023

- 2022 - worked on the ball shooter (the robot ranked #2 nationally). The mechanism was lightweight, included a 2 axis angle adjusting mechanism.
- 2023 - worked on the gripper (#8 nationally) – flexible to enable both cone and cube picking while consistent and robust. The robot finished the second annual competition with 12 wins and 0 losses.
- 2024 - served as the lead CAD mentor, the robot won the Israeli championship qualifying for worlds' championship. In addition, I worked on electronics and fabrication (**3d printing & CNC**).

Skills

CFD (ANSYS Fluent, STAR CCM+) • CAD/FEA (SolidWorks, CREO, ANSYS) • 3D printing & CNC • Topology optimization • Programming • Aerodynamics / Fluid mechanics • Wind Tunnel testing • Thermodynamics• mechatronics

Languages

English: Native level | Hebrew: Full professional proficiency | Arabic: Native

Projects

CFD solver - Built a CFD solver in Python using finite-difference discretization of the Navier–Stokes equations; currently extending the project to a lattice Boltzmann formulation.

3DOF atmospheric reentry trajectory calculator - Built a 3DOF atmospheric reentry trajectory calculator in Python solving the equations of motion over a spherical Earth with aerodynamic forces, heat shield ablation, full dynamical representation of all phases of entry and visualization.